

DJ Library Organizer

User Guide for DJs

Safe migration, genre audit, cleanup review, and Serato/VirtualDJ prep

Important safety note: Always run dry-run and audit first. Use copy mode before move mode. Keep a backup of your music library before moving files. This tool is designed to help organize files, but DJ libraries contain messy metadata, crossover edits, and duplicate versions that should be reviewed.

1. What This Tool Does

- Scans a music library folder and reads audio metadata.
- Detects genres using tags, filenames, folder paths, artist names, and keyword rules.
- Routes tracks into a clean DJ_MUSIC folder structure.
- Detects Clean, Dirty, Intro, Outro, Extended, Acapella, Instrumental, Remix, and similar DJ-use versions.
- Creates CSV reports so you can review before moving files.
- Flags low bitrate, missing metadata, corrupted/tag-error files, duplicates, and Uncategorized tracks.
- Creates M3U playlists for Serato DJ Pro and VirtualDJ import.
- Supports resume mode so large jobs can continue after interruption.

2. Recommended Workflow

Step	Mode	Purpose
1	Audit	Scan the library and create CSV reports. No files move.
2	Dry Run	Preview what the organizer would do for a genre or full library.
3	Copy Test	Copy a small batch into DJ_MUSIC and test in Serato/VDJ.
4	Move by Genre	Move one genre at a time after reviewing reports.
5	Clean Audit	Scan the organized DJ_MUSIC library and find misplaced tracks.
6	Refine Rules	Adjust genre keywords/priority based on what the CSV reveals.

3. Folder Structure

The tool uses a hybrid structure: simple physical folders plus metadata/comments for smart crates and filters.

```
DJ_MUSIC/
├── Hip-Hop/
│   ├── Clean/
│   ├── Dirty/
│   ├── Intro Edits/
│   ├── Acapellas/
│   └── Instrumentals/
├── House/
│   ├── Clean/
│   ├── Dirty/
│   ├── Intro Edits/
│   ├── Acapellas/
│   └── Instrumentals/
├── Reggaeton/
├── Latin/
├── EDM/
├── Videos/
├── Duplicate/
├── Uncategorized/
├── _Reports/
├── _Playlists/
├── _Review_Low_Quality/
├── _Review_Corrupted/
├── _Review_Missing_Tags/
├── _OneTagger_Import/
└── _State/
```

4. Installation

Requirements: macOS, Python 3, and the mutagen Python package.

```
python3 -m pip install mutagen
```

Save the Python file as:

```
/Users/UserName/Documents/Dooms_day/dj_library_organizer_full.py
```

5. Main Commands

Task	Command
Go to project folder	cd "/Users/Name/Documents/Dooms_day"
Audit source library	python3 dj_library_organizer_full.py --audit
Audit first 1000 files	python3 dj_library_organizer_full.py --audit --limit 1000
Dry run Hip-Hop	python3 dj_library_organizer_full.py --dry-run --only "Hip-Hop" --limit 1000 --resume
Safe copy test	python3 dj_library_organizer_full.py --copy --only "Hip-Hop" --limit 1000 --resume
Move one genre	python3 dj_library_organizer_full.py --move --only "Hip-Hop" --resume
Chaos move everything	python3 dj_library_organizer_full.py --move --resume
Clean audit organized library	python3 dj_library_organizer_full.py --clean
Print Bash helper	python3 dj_library_organizer_full.py --bash

6. What Each Mode Means

- **--audit:** Scans the source library and creates genre count/detail CSV reports. No files are moved.
- **--dry-run:** Shows what the organizer would do. No files are copied or moved.
- **--copy:** Copies files into DJ_MUSIC. Originals stay in the source folder.
- **--move:** Moves originals into DJ_MUSIC. Use only after backup and testing.
- **--clean:** Scans the organized DJ_MUSIC folder and creates a cleanup suggestion CSV. No files move.
- **--resume:** Skips files already processed in the same job.
- **--reset-resume:** Clears the resume state for the current job.
- **--limit 1000:** Processes only the first 1000 files for testing.
- **--only "Genre":** Processes only tracks detected as that genre.

7. Understanding the CSV Reports

Reports are saved in:

```
/Users/Username/DJ_MUSIC/_Reports
```

Column	Meaning
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Status	What the script did or would do: WOULD_COPY, COPIED, MOVED, REVIEW, OK, etc.
Detected Genre	The script's best genre guess.
Versions	Clean, Dirty, Intro, Acapella, Instrumental, Remix, etc.
Quality Flag	GOOD, OK_BUT_REVIEW, LOW_BITRATE_REVIEW, UNKNOWN_BITRATE.
Review Flags	Missing artist/title/genre, tag errors, short track review, etc.
Suggested Comment	Suggested comment format for One Tagger/Serato/VDJ.
Current Path	Where the file currently lives.
Suggested Path	Where the tool thinks it belongs.

8. Using the New Library in Serato and VirtualDJ

Serato DJ Pro

1. Open Serato DJ Pro.
2. Go to the Files panel.
3. Browse to /Users/UserName/DJ_MUSIC.
4. Drag genre folders into All or into crates.
5. Analyze files to build waveforms, BPM, and key.
6. Avoid relying on Apple Music/iTunes XML integration for the DJ workflow.

VirtualDJ

7. Open VirtualDJ.
8. Browse to /Users/UserName/DJ_MUSIC.
9. Add folders to favorites or create filter folders.
10. Use metadata/comments for filters after tagging with One Tagger.

9. Best Practices for Large DJ Libraries

- Use genre-by-genre migration when accuracy matters.
- Use chaos move only if your music is fully backed up and you accept that cleanup will be needed.
- Do not delete duplicates immediately. Review them first.
- Do not delete low bitrate tracks until you replace rare edits with better versions.
- Use One Tagger after migration for deeper BPM, key, metadata, and comment cleanup.
- Keep folders simple. Use comments, smart crates, and filters for performance organization.
- Run clean audits regularly after big changes.

10. Troubleshooting

Problem	Fix
SyntaxError: unterminated string literal	A line likely broke during editing. Check the line shown in Terminal and close quotes or parentheses.
CSV is empty	The script may be scanning the wrong source folder, or the file was not saved in VS Code.
Serato smart crates are empty	Tracks must be imported into Serato first. Smart crates only filter Serato's internal database.
Genres are mixed up	Review the CSV, then tune <code>GENRE_KEYWORDS</code> and <code>GENRE_PRIORITY</code> . Open-format libraries require multiple passes.
File name too long	Filename length protection should trim long names; if it still happens, lower <code>MAX_FILENAME_LENGTH</code> .
Command not found / file not found	Run <code>cd</code> into the project folder or use the full path to the script.

11. Suggested GitHub README Summary

DJ Library Organizer

A Python tool for DJs who need to migrate and clean large music libraries outside of Apple Music/iTunes.

Features:

- Audit source libraries with CSV reports
- Detect genre, versions, bitrate, duplicates, and missing tags
- Organize files into a hybrid DJ folder structure
- Support copy, move, dry-run, resume, and clean-audit modes
- Generate M3U playlists for Serato and VirtualDJ
- Create One Tagger-friendly suggestion CSVs

Safe workflow:

1. Run audit
2. Run dry-run
3. Copy a small test batch
4. Test in Serato/VDJ
5. Move one genre at a time
6. Run clean audit and refine rules

Built for open-format DJs managing large, messy libraries. Test small. Back up everything. Trust the CSV.